

# \* K-nearest neighbour classification

query  $\Rightarrow$  Maths = 6, CS = 8  
k = 3

| Maths | CS | Result |
|-------|----|--------|
| 4     | 3  | Fail   |
| 6     | 7  | Pass   |
| 7     | 8  | Pass   |
| 5     | 5  | Fail   |
| 8     | 8  | Pass   |

Euclidean distance  $d = \sqrt{|x_{01} - x_{A1}|^2 + |x_{02} - x_{A2}|^2}$

$$\textcircled{\text{I}} \quad \sqrt{(6-4)^2 + (8-3)^2} = \sqrt{29} = 5.38$$

$$\textcircled{\text{II}} \quad \sqrt{(6-6)^2 + (8-7)^2} = \textcircled{1} \quad \text{Pass}$$

$$\textcircled{\text{III}} \quad \sqrt{(6-7)^2 + (8-8)^2} = \textcircled{1} \quad \text{Pass}$$

$$\textcircled{\text{IV}} \quad \sqrt{(6-5)^2 + (8-5)^2} = \sqrt{10} = 3.16$$

$$\textcircled{\text{V}} \quad \sqrt{(6-8)^2 + (8-8)^2} = \textcircled{2} \quad \text{Pass}$$

Hence Pass